

Multi-Plant Disease Detection and Treatment

Deep Learning-Based Mobile and Web App

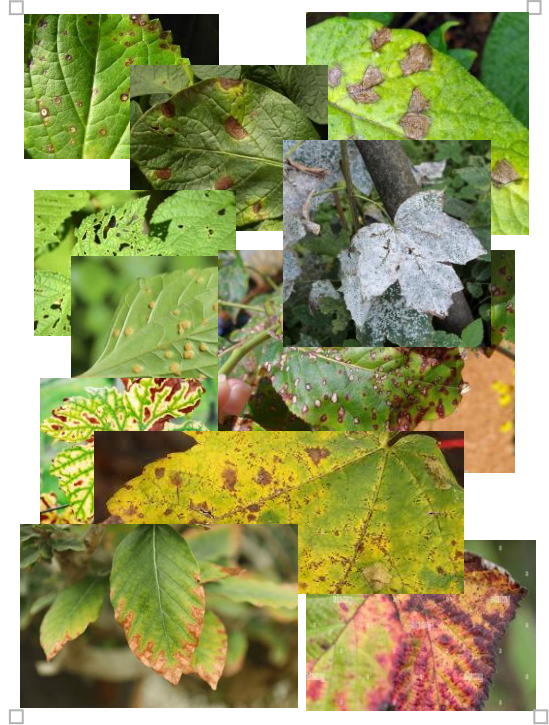
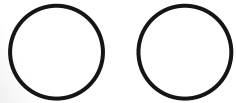


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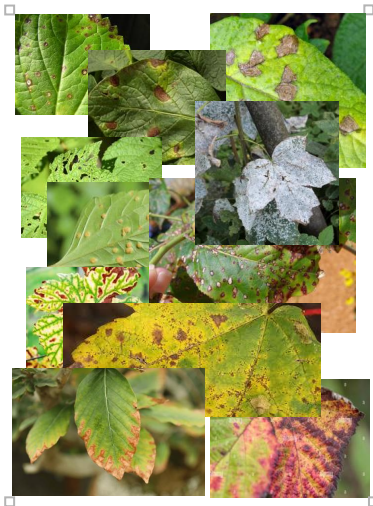
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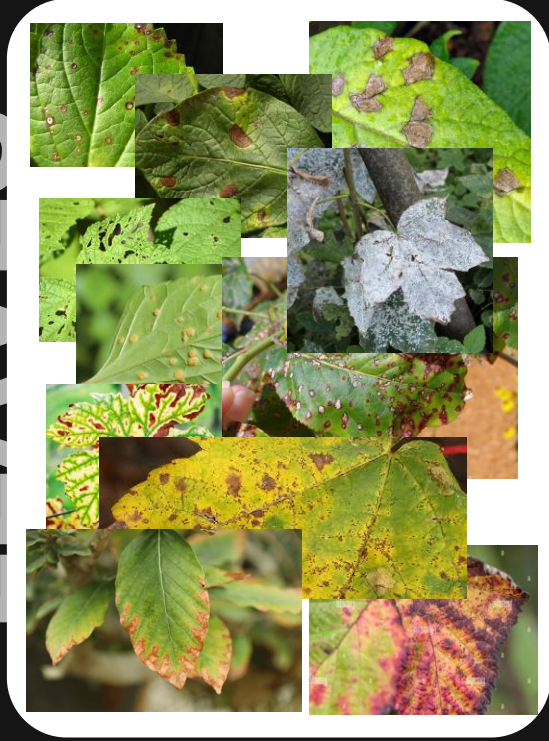
Output of the Mobile and Web App for plan disease detection

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LEAVES



DISEASE

This project is an **AI-powered mobile and web application** that detects plant diseases using **deep learning (CNNs)** and provides **treatment recommendations**, ensuring accurate and accessible disease management for farmers.

PROBLEM STATEMENT

Traditional plant disease detection methods are **manual, time-consuming, and inaccessible to many farmers**, leading to **misdiagnosis and delayed treatment**. Existing digital solutions are often **complex, lack treatment recommendations**, making them difficult for farmers to **use effectively**. This project proposes an **AI-powered mobile and web application** using **deep learning (CNNs)** to provide **real-time, accurate disease detection and treatment recommendations**, ensuring accessibility for farmers through a **user-friendly, multilingual interface**.

OBJECTIVES



**Disease Detection and
Classification**



**Real-Time Analysis
and Scalability**



**Actionable
Recommendations and
Sustainable Practices**



**Multi-Language
Support**

PROPOSED SOLUTION



**Deep Learning-Based
Disease Detection**



**Mobile & Web App for
Easy Accessibility**



Cloud-Based Backend



Multilingual Support



**Actionable Treatment
Recommendations**

METHODOLOGY

**Data Collection &
Preprocessing**

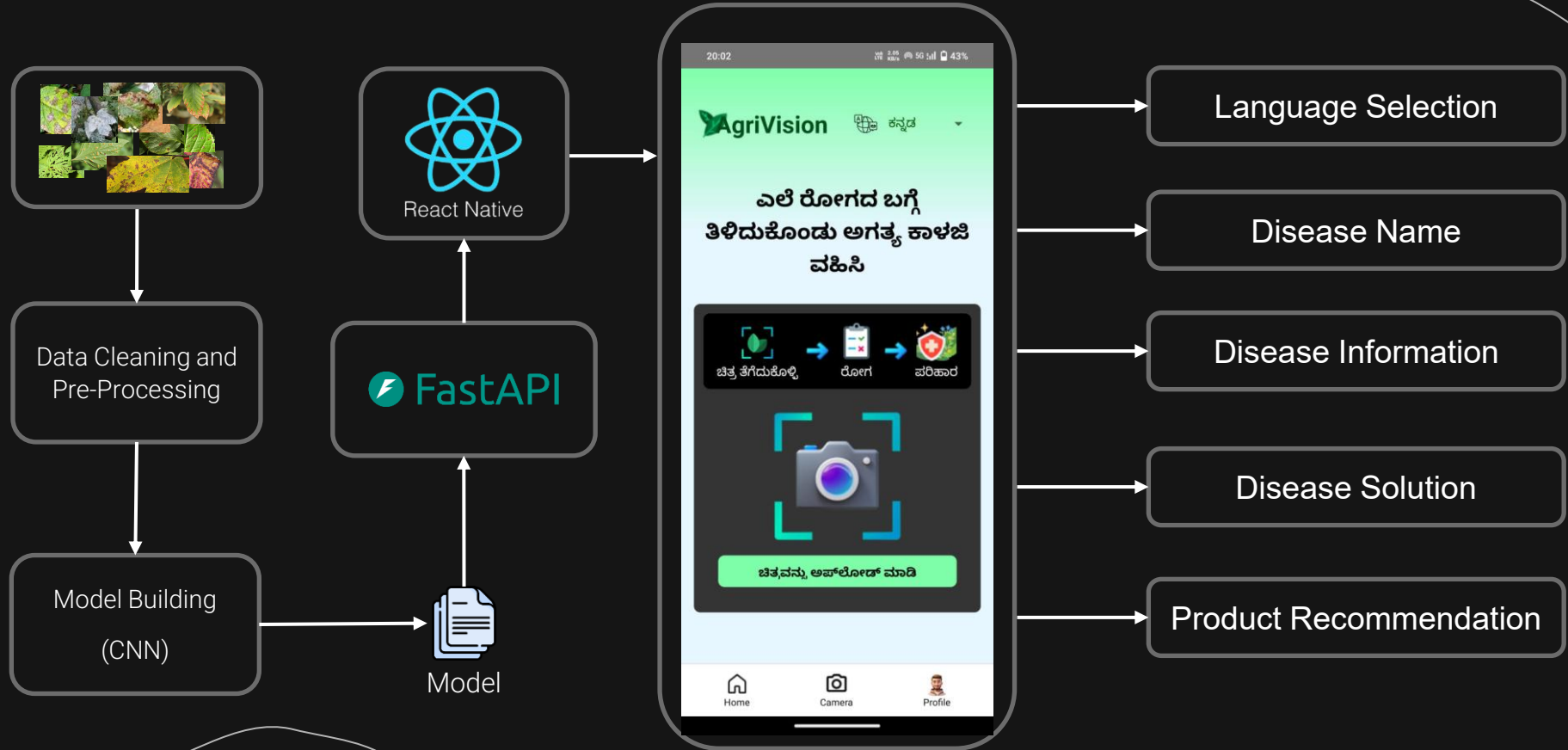
**Model Development &
Training**

**System Architecture &
Deployment**

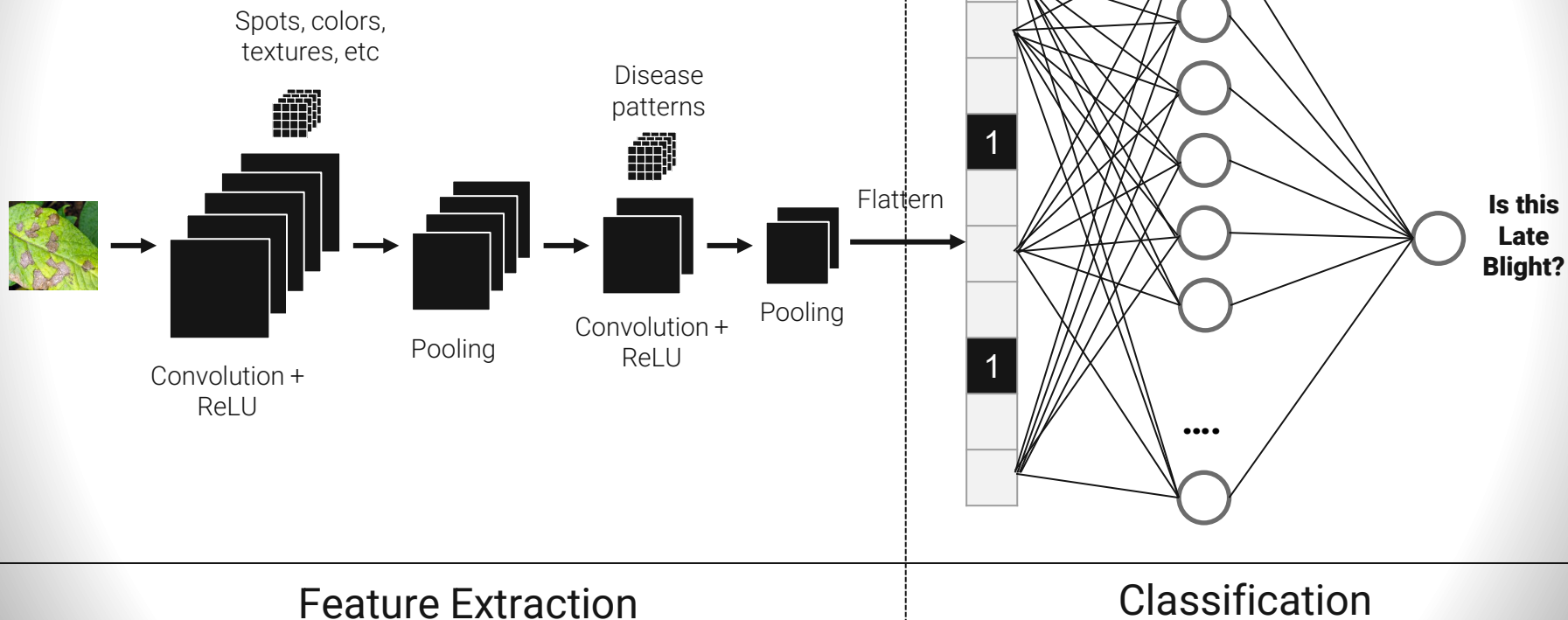
**User Interaction &
Output Generation**

**Performance Evaluation
& Testing**

SYSTEM ARCHITECTURE



CNN ARCHITECTURE



OUTPUT

19:54 5G 100% 45%



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Select a language ▼

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Confirm Your Password

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Have an Account? Log in

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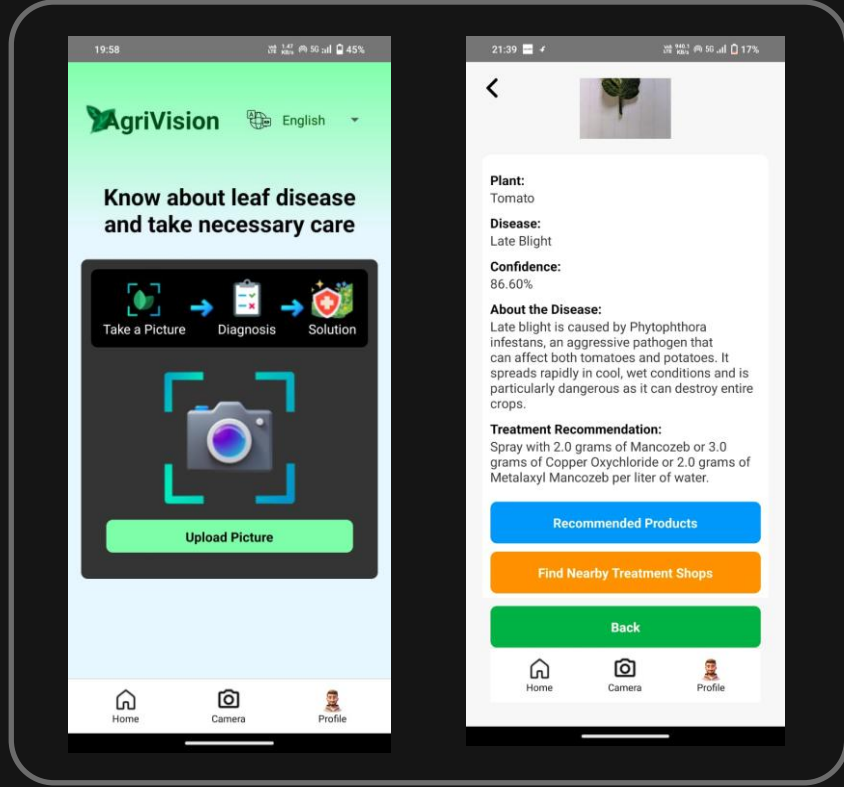
App Info

Sign Out

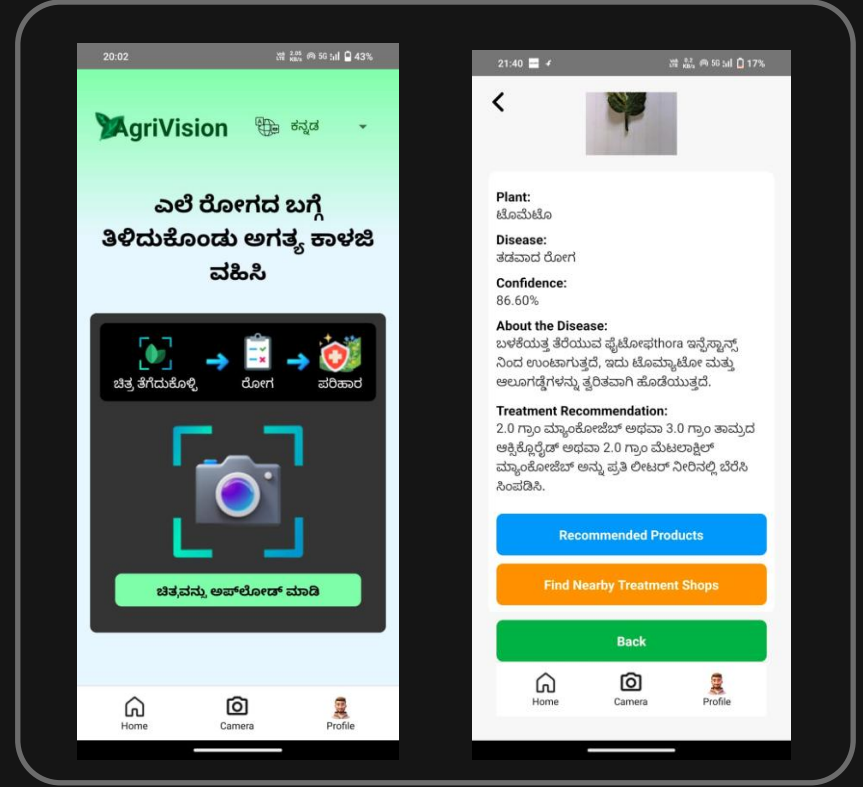
Share

Home Camera Profile

OUTPUT



ENGLISH



ಕನ್ನಡ

OUTPUT

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Detected Disease



plantName: Potato

diseaseName: Early Blight

Confidence: 1.00%

aboutDisease: Early blight is caused by the fungus *Alternaria solani*. It thrives in warm, moist conditions and can be spread by wind, rain, and contaminated tools.

treatment: Apply fungicides and practice crop rotation to reduce disease impact.

Recommended Products

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English

Know about leaf disease and take necessary care



Take a Picture



Diagnosis



Solution



Upload Picture

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Software Requirements

Backend & AI Model

- Python
- FastAPI
- TensorFlow/Keras
- MongoDB
- AWS EC2

Frontend

- React Native
- ReactJS

Development & Testing Tools

- Postman
- Jupyter Notebook
- Android Studio

Hardware Requirements

Development Environment

- Intel i5 (8th Gen) / AMD Ryzen 5
- 16GB RAM
- 512GB SSD

Server-Side Requirements

- AWS EC2

Client-Side (End User Devices):

- Android (v8.0+)
- iOS (v13+)
- Chrome
- Firefox
- Edge
- Safari

Applications



**Agriculture and
Farming**



**Agri tech
Solutions**



E-Commerce



Educational Tool

CONCLUSION

This project provides an innovative solution for early plant disease detection and management, leveraging deep learning and accessible through mobile and web platforms. With key components like data processing, model development, and mobile app functionality already completed, the project is on track to support farmers and agricultural stakeholders in safeguarding crop health and optimizing yield through timely disease prediction and treatment recommendations.

REFERENCES

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THANK



YOU